

Name: _____

Answer all questions and turn in before leaving class today. Include your work for partial credit.

Assume the following :

- I have 3 queues : Q0 has highest priority, Q1 has next priority, and Q2 has lowest priority.
- When a job enters the system, it starts in the highest priority queue, Q0.
- Job A is a long-running CPU-intensive job.
- Job B arrives at time step $T=100$ and is an interactive job that requires 10ms of CPU time followed by 10ms of I/O.
- The time slice is 20ms.
- Job A will run on the CPU while job B is performing I/O.

1. Assume each job moves down a priority only if it uses up its entire time slice before switching to I/O. Also, assume there is no priority boost.

(a) At $T = 150\text{ms}$, what queue is job A in?

(b) At $T = 150\text{ms}$, what queue is job B in?

2. Assume the scheduler moves a job down one priority as soon as it runs on the CPU for a 'time slice' amount of time, regardless whether the entire time slice is used at one time. Again, there is no priority boost. At time = 150ms, what queue is job B in?

3. Assume the scheduler moves a job down one priority as soon as it runs on the CPU for a 'time slice' amount of time, regardless whether the entire time slice is used at one time. Assume there is a priority boost every 50ms. At $T = 140\text{ms}$, what queue is job A in?